Using a Monte Carlo Simulation to model celestial FRBs

Ethan Thompson★
Department of Physics and Astronomy, West Virginia University, Morgantown, WV 26501, USA

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ABSTRACT

Fast radio bursts (FRBs) are one of the most interesting phenomena that is studied today. Modeling these FRBs is a means of exploring the likely populations of these energetic events while constraining the possible possibilities revolving around their creation. I present a simple Monte Carlo simulation of the FRB population which aims to replicate the flux vs. dispersion measure (DM) of the entire population. I find it needed that the population of all modeled FRBs and host galaxies with dispersion measure (DM) are uniformly distributed at a distance of 0 to 5 Gpc and are modeled by a normal distribution with a mean of 70 and a standard deviation of 15 cm^{-3} pc. The FRB luminosity distribution is drawn from a log-normal distribution which has a mean of 0.1 and a standard deviation of 0.8. This simulation could be improved by using a more realistic data for the dispersion measure in the form of the actual FRB distances rather than randomly generated ones.

1 INTRODUCTION

The first fast radio burst (FRB) was found by Lorimer et al. (2007). FRBs are millisecond duration radio pulses that are emitted from celestial sources. Even though the flux densities of FRBs are similar to already known objects like pulsars, their distances are several million times greater, implying that their luminosities are around 10^{12} times those of regular pulsars. This led to the conclusion that they are new celestial phenomena (Zhang 2020). Instrumental and computational advances in astronomy, primarily in the Canadian Hydrogen Intensity Mapping Experiment (CHIME) project (CHIME/FRB Collaboration et al. 2021), have led to the discovery of more than 600 FRBs today.

In this paper, I am using a Monte Carlo simulation to model FRBs. I will model the FRB population using the flux density–dispersion measure plane. In section 2 I describe the simulation, in section 3 I giving my results, in section 4 I discuss the implications of these before concluding in section 5.

2 METHODS

2.1 Simulation

To ensure that the distribution of FRBs has a constant density per unit comoving volume out to distance $D_{\rm max}$ (max distance considered in simulation), I set up the simulation as follows. Given a random deviate r in the range $0 < r < 1$, the distance to each FRB

$$D = D_{\rm max}\sqrt[3]{r}, \tag{1}$$

where D is the comoving distance. Next, I convert these distances to redshifts making use of the standard relationship (see, e.g., Hogg 1999) where

$$D(z) = \frac{c}{H_0}\int_0^z \frac{{\rm d}z'}{\sqrt{\Omega_m(1+z')^3+\Omega_\Delta}}, \tag{2}$$

where c is the speed of light, H_0 is the Hubble constant, and Ω_m and Ω_Λ are the total energy densities of matter and dark energy, respectively. For this study, I adopted $H_0 = 68$ km s^{-1} Mpc^{-1}, $\Omega_m = 0.32$ and $\Omega_\Lambda = 0.68$. Dispersion measure (DM) is one of the primary observable quantities for FRBs. To model the observed DM values, $\rm DM_{obs}$ I add the various contributions together as follows:

$$\rm DM_{obs} = DM_{MW} + DM_{IGM} + \frac{DM_{host}}{(1+z)}. \tag{3}$$

In this expression, $\rm DM_{MW}$ is the contribution to DM from electrons in our galaxy, $\rm DM_{IGM}$ is the contribution to DM from electrons in the intergalactic medium, and $\rm DM_{host}$ is the contribution of electrons to DM from the host galaxy.

To model the flux density–DM distribution, for each model FRB, I compute the flux density

$$F = \frac{L}{4\pi(1+z)^2 D^2}, \tag{4}$$

where D is the comoving distance defined above and L is the luminosity of each FRB. I chose luminosity values from a log-normal distribution. This introduces two additional model parameters: the mean, μ_L, and the standard deviation, σ_L. In practice, μ_L and σ_L are used to generate the base-10 logarithm of L, where L is assumed to be in units of Jy Gpc^2.

3 RESULTS

The results are shown in figure 1 for the set of the model FRBs that were above the 0.2 Jy limit. Also shown in the figure is the known FRBs by the CHIME/FRB catalog. The model parameters are adjusted to be in agreement with the model and true data. These parameters are: D_{max} = 6 Gpc, μ_L = 1.0, σ_L = 0.8, μ_{host} = 70 cm^{-3} pc, and σ_{host} = 15 cm^{-3} pc.

4 DISCUSSION

This model only touches the tip of the iceberg. Due to the parameters and simplification set at the beginning of the process, a great percentage of FRBs are not represented in this model. Compared to

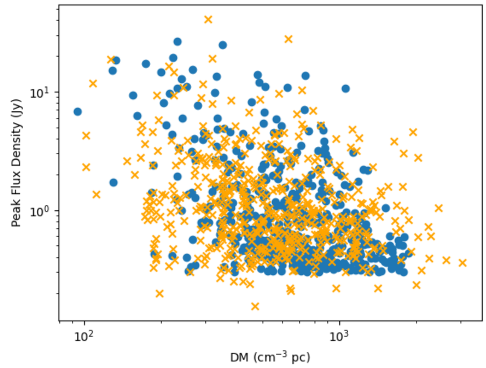

Figure 1. This is the final Flux vs.DM plot. The blue markers are the model FRBs and the orange markers are the real FRBs from CHIME.

other models, this one had simplifications which eventually lead to the plot in section 3.

4.1 Simplification and Parameter changes

This plot and model is very elementary when compared to other models. Thus, there was simplification, including : (i) assuming they are uniformly distributed with a distribution of luminosities in space:(ii) assuming DM_{IGM} = 1000z: (iii) no consideration for repeating sources:(iv) making an oversimplified model of the DM for the Milky Way and intergalactic medium. When first making the flux verses dispersion measure plot, I noticed some problems with the plot. The values were all over the place and it didn't show a true correlation, so parameter changes were implemented. As seen in section 3, D_{max} was equal to 5 Gpc. This was the parameter until the CHIME data and the model were put together. To better fit the data and CHIME, D_{max} was changed to D_{max} = 6 Gpc. The other parameters that were adjusted are $\mu_L = 1.0$, $\sigma_L = 0.8$, $\mu_{host} = 70$ cm^{-3} pc, and $\sigma_{host} = 15$ cm^{-3} pc.

4.2 Correlation between Model and CHIME data

After the parameter changes, the CHIME data closely matched the model data. The same general shape on the plot was produced. Of course, the CHIME data does not align perfectly with all of the data. However, for the model of this level with the parameters put with this model, it does a good job.

4.3 How to make more realistic?

This model runs of randomly generated distances from 0-5 Gpc. However, to make more realistic, there is a few things we can do. One idea is that instead of randomly generated distances, we can use a set of distances that closely relate to real FRB distances. This will get us more accurate redshift and dispersion measures, thus improving the quality of this model. Secondly, we can use a more powerful program and computer system that does not need simplifications to make the calculations easier, thus making the program realistic.

5 CONCLUSIONS

Even though there was simplifications, this is a pretty good model for FRB distribution in flux density - dispersion measure space for the CHIME data. I have found that a plot of flux density vs. dispersion measure is compatible with known FRBs today. Using this finding, I see good evidence that a log-normal form is indeed true for the flux luminosities of FRBs. Due to the simplifications above, future work in this area would be highly recommended. I would love to see a project that takes this framework and makes it into a model that predicts FRBs at a high success rate. As technology advances, especially in the realms of computing and AI, hopefully we can see something like this in the future. **? ? ?**

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